第一章：绪论

概览

- 机器学习中有哪些基本的概念？
- 机器学习的一般流程？由哪些部分组成？
- 有哪些机器学习的方法？可以分成几类？代表方法是什么？
- 不同机器学习方法的统一框架？

大纲

- 引言
- 基本术语
- 假设空间
- 归纳偏好
- 发展历程
- 应用现状
- 阅读材料

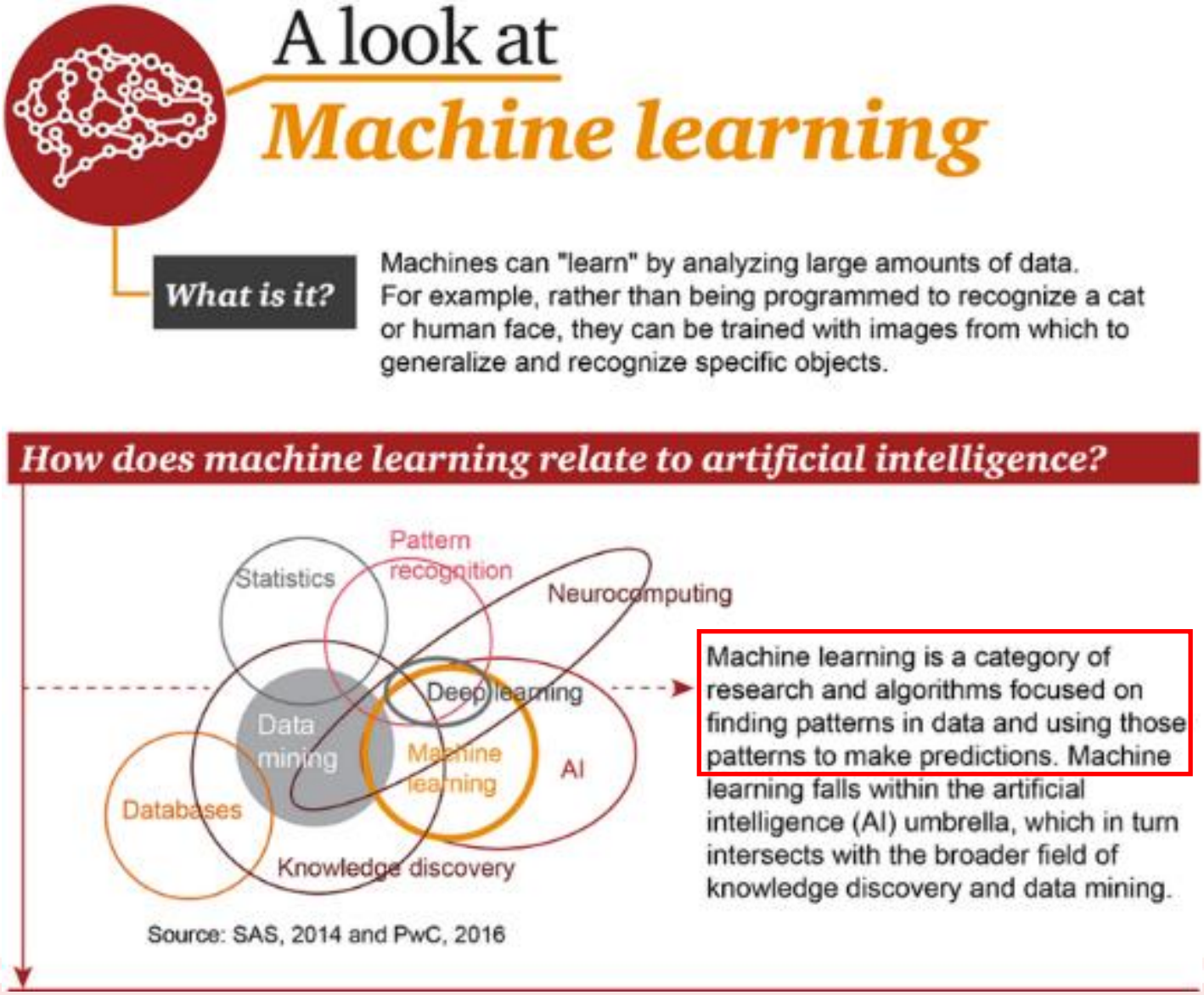
机器学习

“假设用***P***来评估计算机程序在某任务类***T***上的性能，若一个程序通过利用经验***E***在***T***中任务上获得了性能改善，则我们就说关于***T***和***P***，该程序对***E***进行了学习”

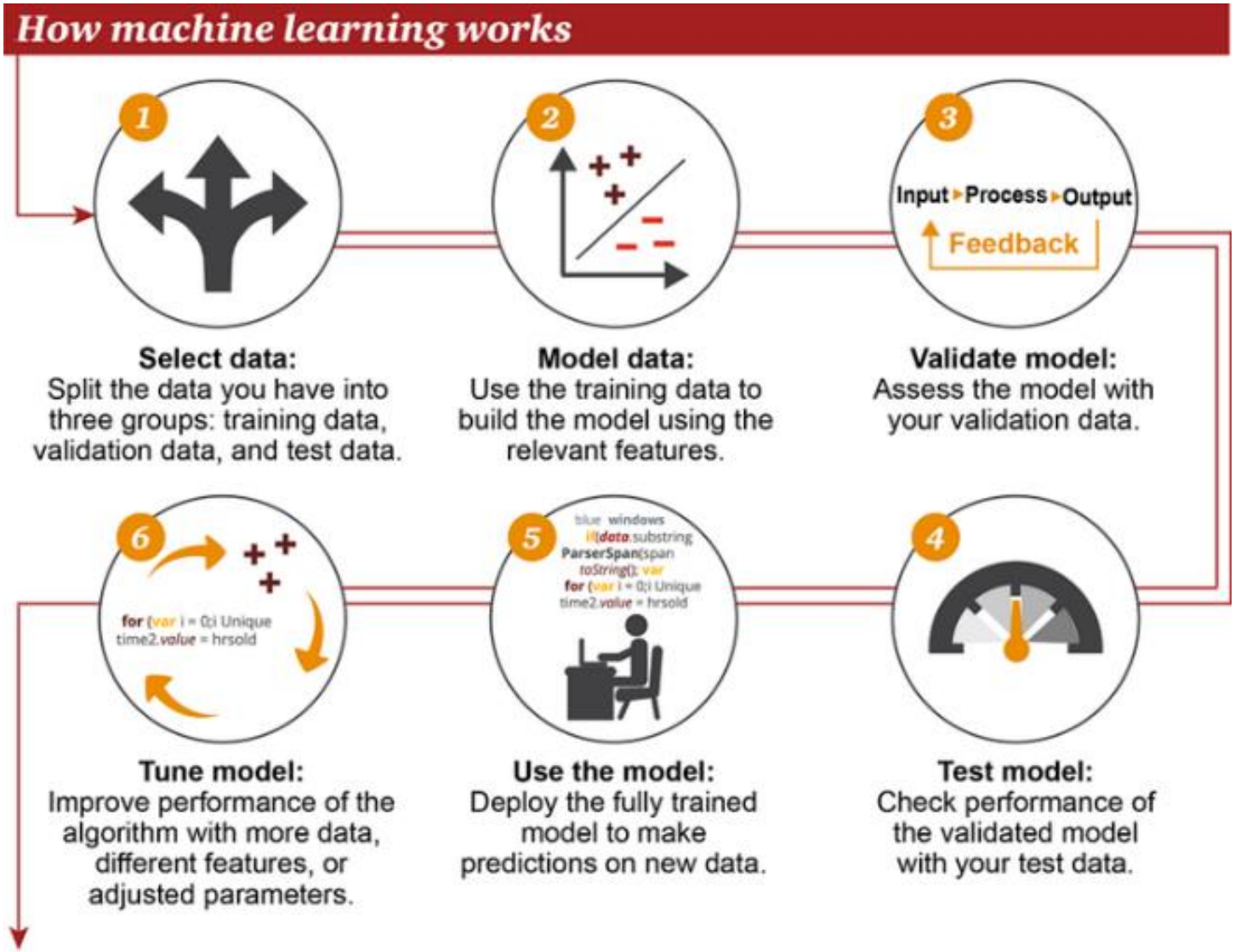
“A computer program is said to *learn* from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.”

机器学习致力于研究如何通过计算的手段，利用经验来改善系统自身的性能，从而在计算机上从数据中产生“模型”，用于对新的情况给出判断。

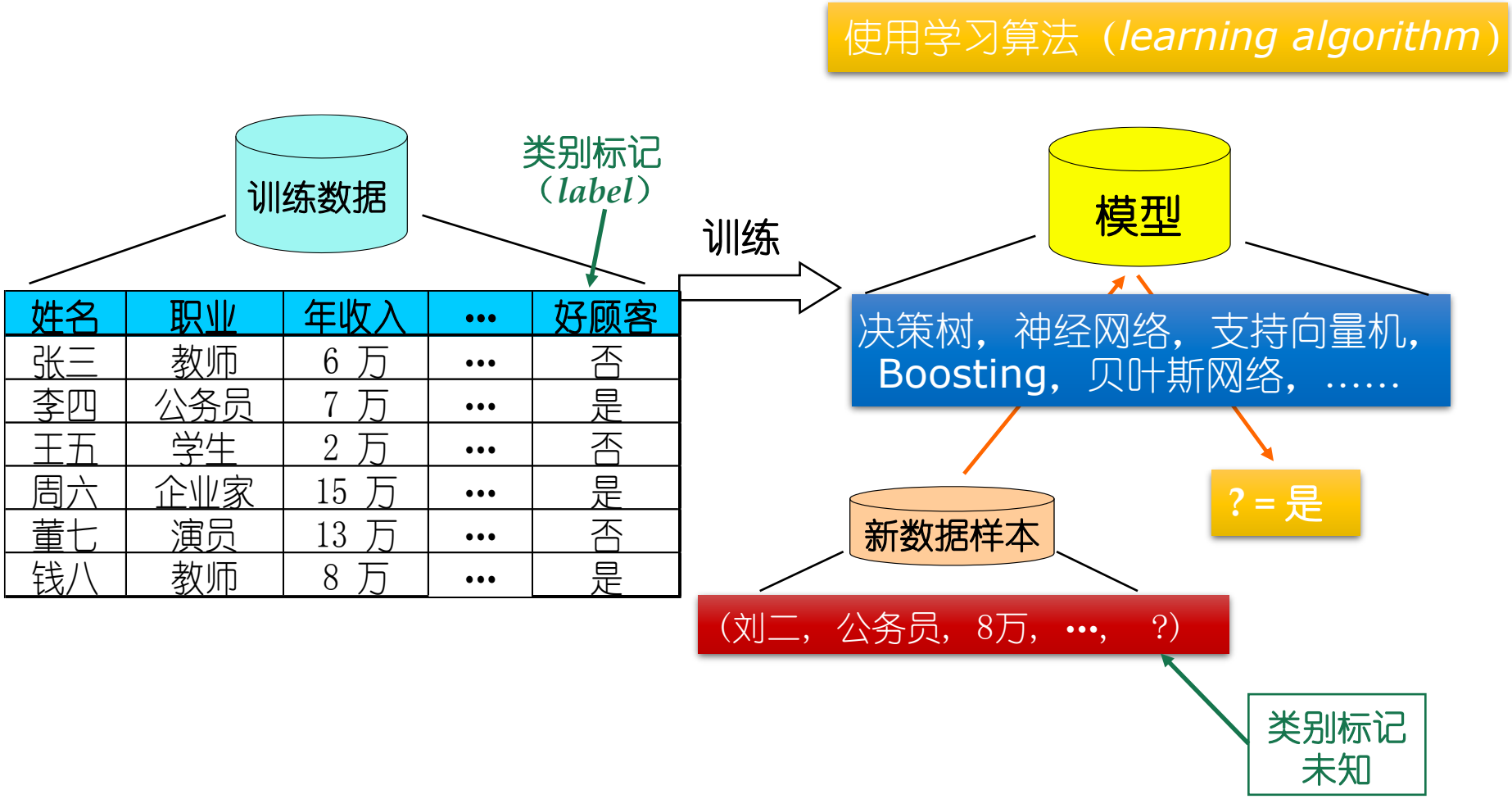
机器学习与AI


A look at
Machine learning
What is it?
Machines can "learn" by analyzing large amounts of data. For example, rather than being programmed to recognize a cat or human face, they can be trained with images from which to generalize and recognize specific objects.
How does machine learning relate to artificial intelligence?
Statistics
Pattern recognition
Neurocomputing
Deep learning
Data mining
Machine learning
AI
Databases
Knowledge discovery
Machine learning is a category of research and algorithms focused on finding patterns in data and using those patterns to make predictions. Machine learning falls within the artificial intelligence (AI) umbrella, which in turn intersects with the broader field of knowledge discovery and data mining.
Source: SAS, 2014 and PwC, 2016

典型的机器学习过程


How machine learning works
1
2
3
Input ▸ Process ▸ Output
Feedback
Select data:
Split the data you have into three groups: training data, validation data, and test data.
Model data:
Use the training data to build the model using the relevant features.
Validate model:
Assess the model with your validation data.
6
5
4
Tune model:
Improve performance of the algorithm with more data, different features, or adjusted parameters.
Use the model:
Deploy the fully trained model to make predictions on new data.
Test model:
Check performance of the validated model with your test data.

典型的机器学习过程



姓名	职业	年收入	…	好顾客
张三	教师	6 万	…	否
李四	公务员	7 万	…	是
王五	学生	2 万	…	否
周六	企业家	15 万	…	是
董七	演员	13 万	…	否
钱八	教师	8 万	…	是

机器学习

- Machine Learning = LAMBDA
  - L: Loss
  - A: Algorithm
  - M: Model
  - BD: Big Data
  - A: Application

大纲

- 引言
- 基本术语
- 假设空间
- 归纳偏好
- 发展历程
- 应用现状
- 阅读材料

基本术语-数据

特征　　标签

编号	色泽	根蒂	敲声	好瓜
1	青绿	蜷缩	浊响	是
2	乌黑	蜷缩	沉闷	是
3	青绿	硬挺	清脆	否
4	乌黑	稍蜷	沉闷	否
1	青绿	蜷缩	沉闷	?

训练集

测试集

d维特征向量$x=(x_1,x_2,\ldots,x_d)$　特征空间

基本术语-任务

- 有无标记信息
  - 监督学习：训练数据包含标记。算法学习从特征到标记的映射。典型任务包括**分类**和**回归**。
  - 无监督学习：训练数据无标记。算法自动发现数据中的结构。典型任务是**聚类**（clustering），此外还有降维、异常检测等。
  - 半监督学习：结合少量标记数据和大量无标记数据，利用无标记数据来提升学习性能。这在现实应用中非常常见，因为标记数据往往昂贵，而无标记数据容易获取。

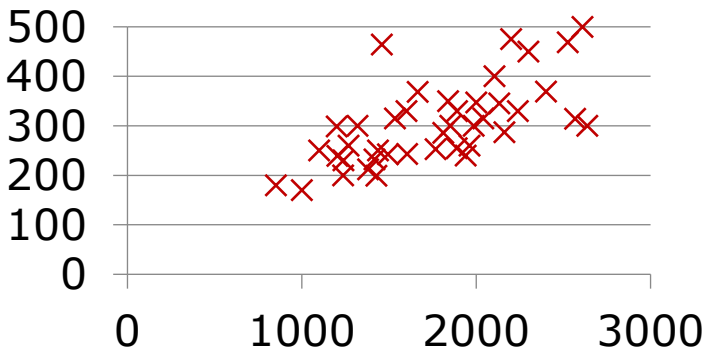
基本术语-任务

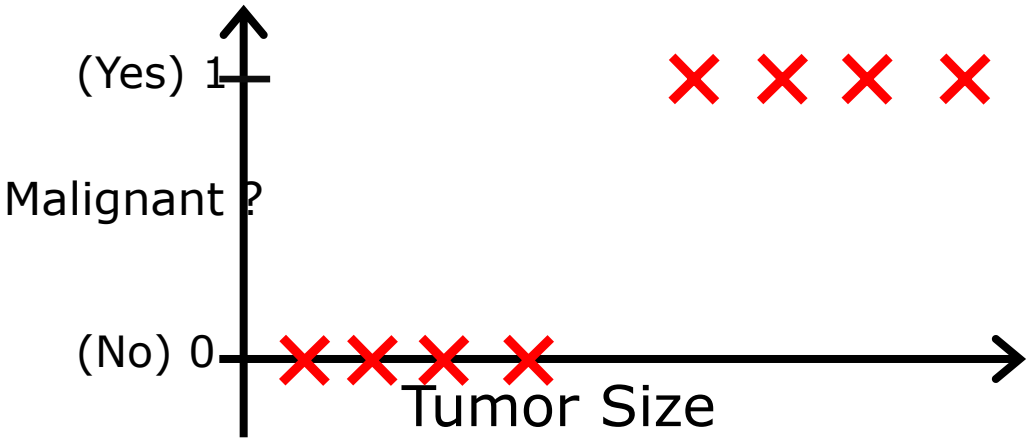
- 预测目标：
  - 分类:离散值
    - 二分类:好瓜;坏瓜
    - 多分类:冬瓜;南瓜;西瓜
  - 回归:连续值

    瓜的成熟度/瓜的价格
  - 聚类:无标记信息

    按照某些特征把物体归为一类，例如绿皮、红心的都是西瓜

Supervised Learning-回归 VS 分类





基本术语-泛化能力

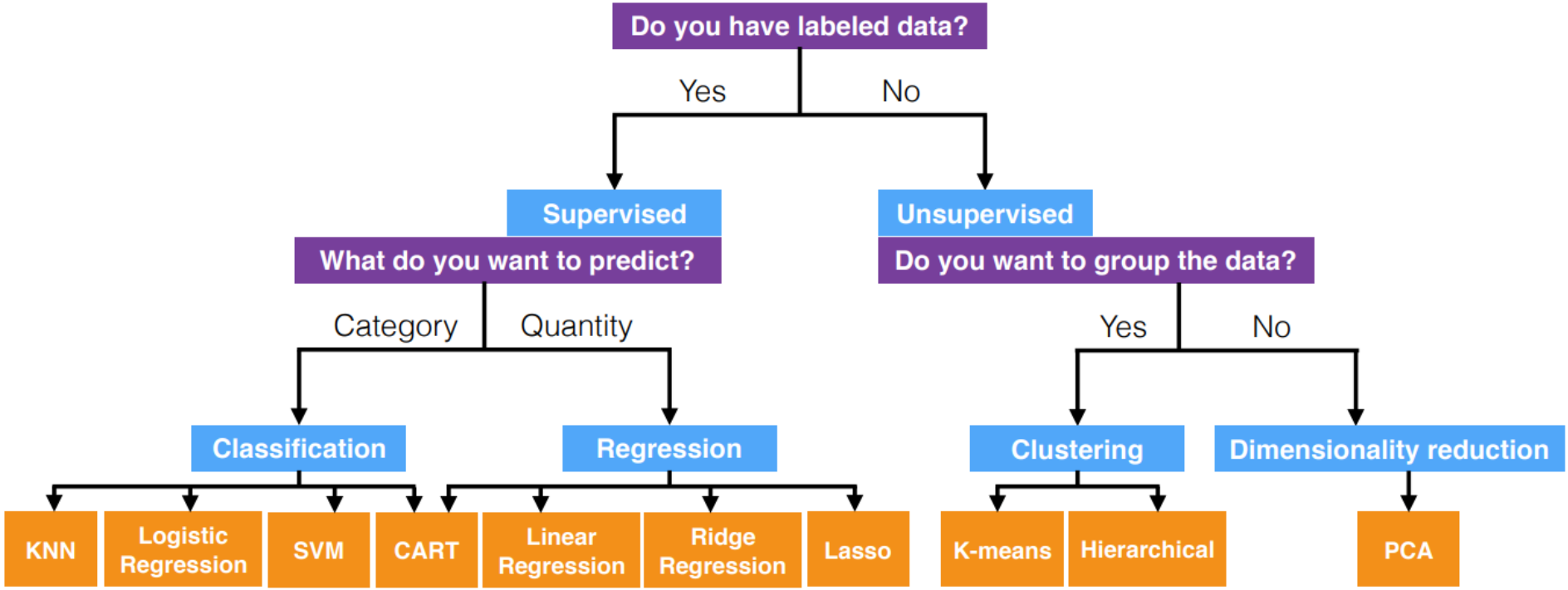
机器学习的目标是使得学到的模型能很好的适用于“新样本”，而不仅仅是训练集合，我们称模型适用于新样本的能力为泛化(generalization)能力，也叫做举一反三的能力。

3+3=6 → 8+9=17

通常假设样本空间中的样本服从一个未知分布 $\mathcal{D}$,样本从这个分布中独立获得，即“独立同分布”(i.i.d)。一般而言训练样本越多越有可能通过学习获得强泛化能力的模型

机器学习算法

Machine Learning Methods



CME 250: Introduction to Machine Learning, Winter 2019 3

cme250_lecture7 (stanford.edu)

大纲

- 引言
- 基本术语
- 假设空间
- 归纳偏好
- 发展历程
- 应用现状
- 阅读材料

假设空间

假设空间：监督学习（supervised learning）的任务是学习一个模型，使模型能够对任意给定的输入，对其相应的输出做出一个好的预测。模型属于由输入空间到输出空间的映射的集合，这个集合就是假设空间(hypothesis space)。我们也可以将学习过程看作一个在所有假设组成的空间中进行搜索的过程，搜索目标是找到与训练集"匹配"的假设，即能够将训练集中的瓜判断正确的假设。

假设空间

编号	色泽	根蒂	敲声	好瓜
1	青绿	蜷缩	浊响	是
2	乌黑	蜷缩	沉闷	是
3	青绿	硬挺	清脆	否
4	乌黑	稍蜷	沉闷	否

假设空间：(色泽=?)∧(根蒂=?)∧(敲声=?)↔好瓜

色泽（3种）：青绿+乌黑+什么值都合适的情况；

根蒂（4种）：蜷缩+硬挺+稍蜷+什么值都合适的情况；

敲声（4种）：浊响+清脆+沉闷+什么值都合适的情况；

还有一种假设组成——可能"好瓜"这个概念就不成立，用∅表示这种假设。

假设空间大小：3*4*4+1=49

假设空间

编号	色泽	根蒂	敲声	好瓜
1	青绿	蜷缩	浊响	是
2	乌黑	蜷缩	沉闷	是
3	青绿	硬挺	清脆	否
4	乌黑	稍蜷	沉闷	否

49个假设空间（9个示例）

(1) (色泽=青绿) ^ (根蒂=蜷缩) ^ (敲声=浊响)

(2) (色泽=青绿) ^ (根蒂=蜷缩) ^ (敲声=清脆)

(3) (色泽=青绿) ^ (根蒂=蜷缩) ^ (敲声=沉闷)

(4) (色泽=青绿) ^ (根蒂=硬挺) ^ (敲声=浊响)

(5) (色泽=青绿) ^ (根蒂=硬挺) ^ (敲声=清脆)

(6) (色泽=青绿) ^ (根蒂=硬挺) ^ (敲声=沉闷)

(7) (色泽=青绿) ^ (根蒂=稍蜷) ^ (敲声=浊响)

(8) (色泽=青绿) ^ (根蒂=稍蜷) ^ (敲声=清脆)

(9) (色泽=青绿) ^ (根蒂=稍蜷) ^ (敲声=沉闷)

⋮

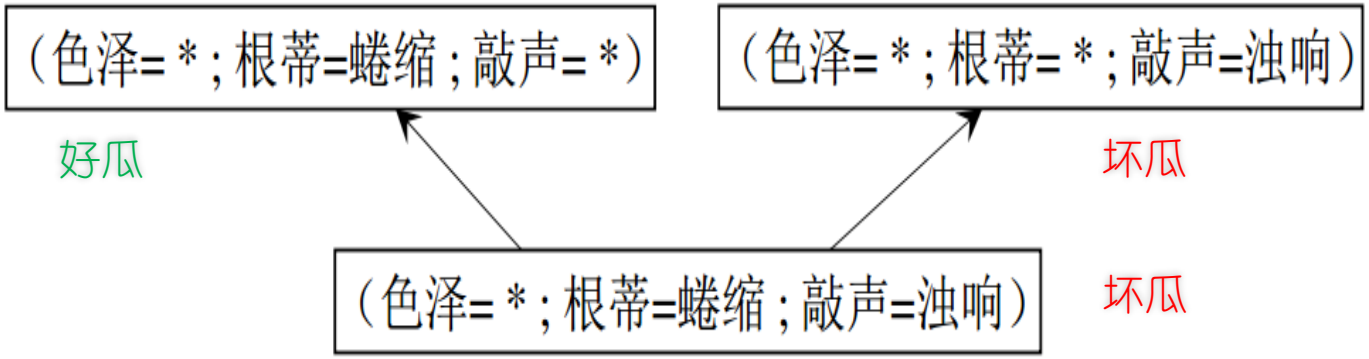
假设空间：(色泽=?)∧(根蒂=?)∧(敲声=?)↔好瓜

大纲

- 引言
- 基本术语
- 假设空间
- 归纳偏好
- 发展历程
- 应用现状
- 阅读材料

归纳偏好

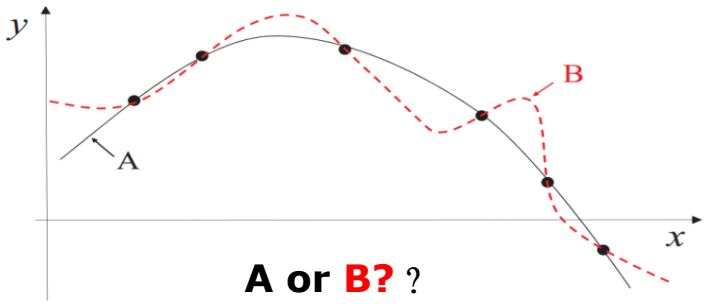
假设空间中有三个与训练集一致的假设，但他们对（色泽=青绿；根蒂=蜷缩；敲声=沉闷）的瓜会预测出不同的结果：



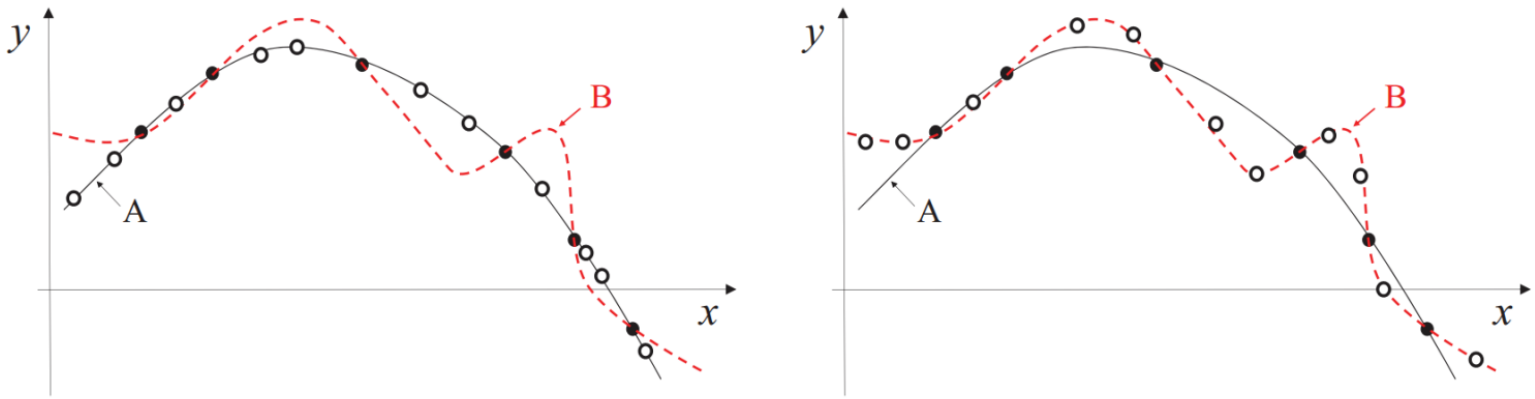
选取哪个假设作为学习模型？

归纳偏好

学习过程中对某种类型假设的偏好称作归纳偏好



存在多条曲线与有限样本训练集一致



(a) A 优于 B　　(b) B 优于 A

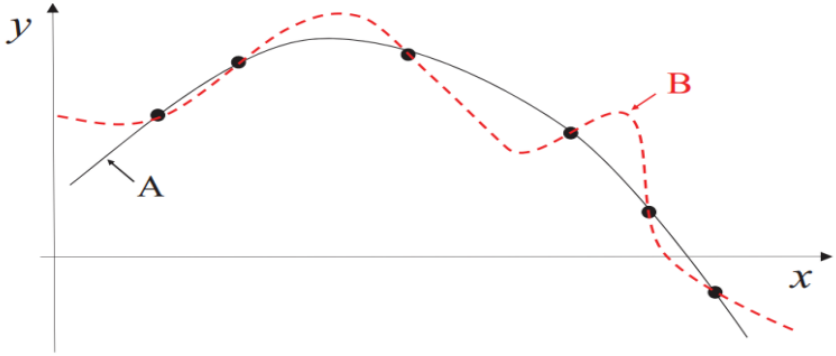
没有免费的午餐. (黑点: 训练样本; 白点: 测试样本)

归纳偏好

归纳偏好可看作学习算法自身在一个可能很庞大的假设空间中对假设进行选择的启发式或“价值观”.

“奥卡姆剃刀”是一种常用的、自然科学研究中最基本的原则，即“若有多个假设与观察一致，选最简单的那个”.

奥卡姆剃刀也会有不适用，当无法判断哪个更简单时，需要其他机制来解决这个问题



存在多条曲线与有限样本训练集一致

1. 好的配偶=体型任意+富有+性别任意

2. 好的配偶=体型好+财富任意+异性

3. 好的配偶=体型任意+富有+异性

2和3无法区分哪个更好？

大纲

- 引言
- 基本术语
- 假设空间
- 归纳偏好
- 发展历程
- 应用现状
- 阅读材料

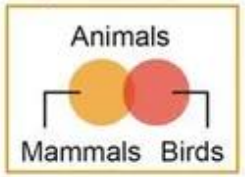
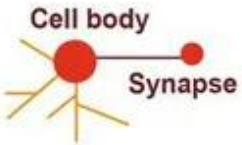
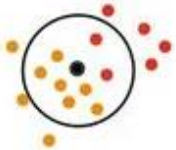
五大流派

A look at
Machine learning evolution

Overview

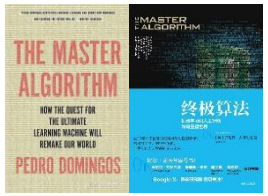
For decades, individual "tribes" of artificial intelligence researchers have vied with one another for dominance. Is the time ripe now for tribes to collaborate? They may be forced to, as collaboration and algorithm blending are the only ways to reach true artificial general intelligence (AGI). Here's a look back at how machine learning methods have evolved and what the future may look like.

What are the five tribes?

Symbolists	Bayesians	Connectionists	Evolutionaries	Analogizers
				
Use symbols, rules, and logic to represent knowledge and draw logical inference	Assess the likelihood of occurrence for probabilistic inference	Recognize and generalize patterns dynamically with matrices of probabilistic, weighted neurons	Generate variations and then assess the fitness of each for a given purpose	Optimize a function in light of constraints ("going as high as you can while staying on the road")
Favored algorithm Rules and decision trees	**Favored algorithm** Naive Bayes or Markov	**Favored algorithm** Neural networks	**Favored algorithm** Genetic programs	**Favored algorithm** Support vectors

Source: Pedro Domingos, *The Master Algorithm*, 2015

知乎 @Gang Tao



1 符号主义

Deduction

Ibn al Haytam is a muslim
\+ Muslims beleive in god

= ?

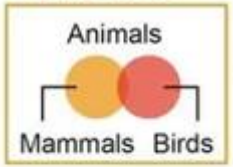
Induction

Ibn al Haytam is a muslim
\+ ?

= Ibn al haytal beleives in god

知乎 @Gang Tao

Symbolists

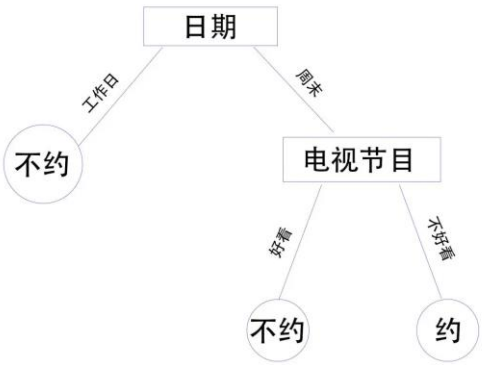


Use symbols, rules, and logic to represent knowledge and draw logical inference

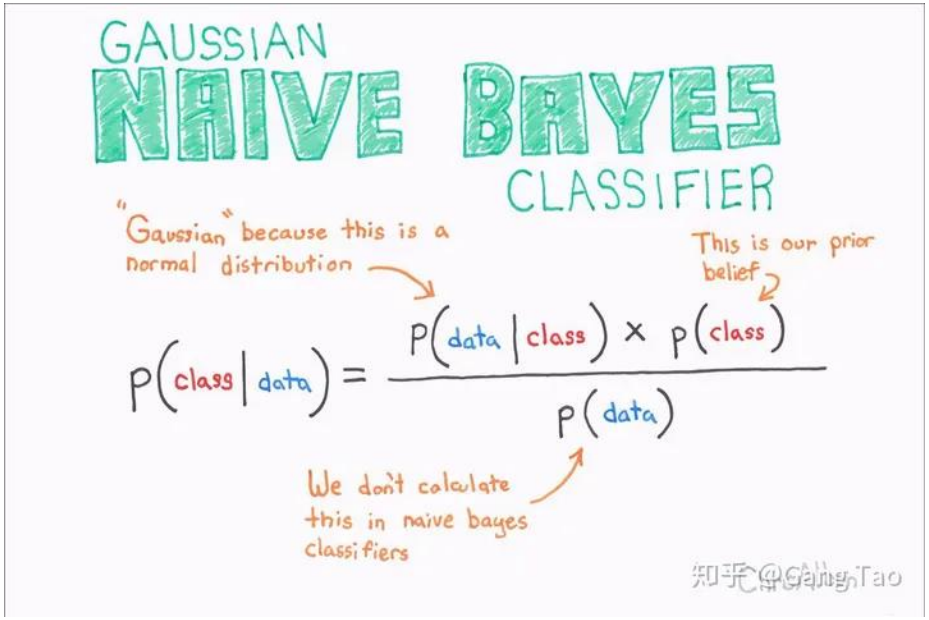
Favored algorithm
Rules and decision trees

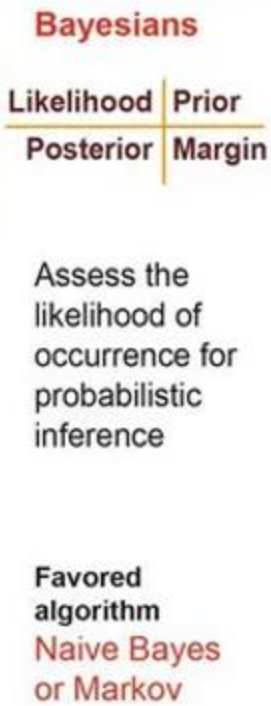
归纳推理

序号	星期几	约会方式	天气	晚间电视节目	是否答应约会
1	周末	吃饭	暖	不好看	否
2	周末	泡吧	暖	不好看	是
3	周末	泡吧	暖	不好看	是
4	周末	泡吧	冷	好看	否
5	周末	泡吧	冷	不好看	?

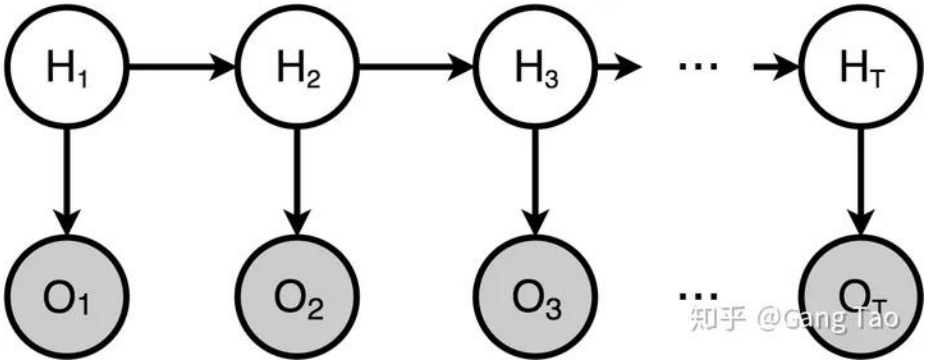


2 贝叶斯主义


GAUSSIAN
NAIVE BAYES
CLASSIFIER
"Gaussian" because this is a normal distribution
This is our prior belief
$P(\text{class} \mid \text{data}) = \frac{P(\text{data} \mid \text{class}) \times P(\text{class})}{P(\text{data})}$
We don't calculate this in naive bayes classifiers
知乎 @Gang Tao

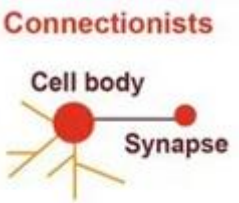

Bayesians
Likelihood
Prior
Posterior
Margin
Assess the likelihood of occurrence for probabilistic inference
Favored algorithm
Naive Bayes or Markov

——概率推理


H_1
H_2
H_3
…
H_T
O_1
O_2
O_3
…
O_T
知乎 @Gang Tao

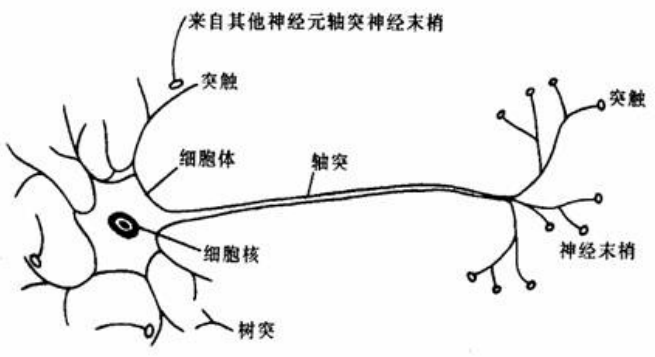
E
P(H/E)
H

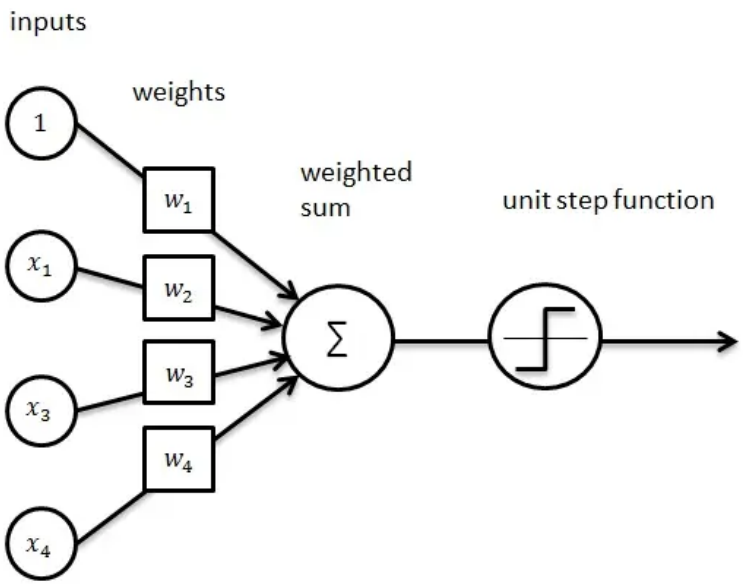
3 连接主义



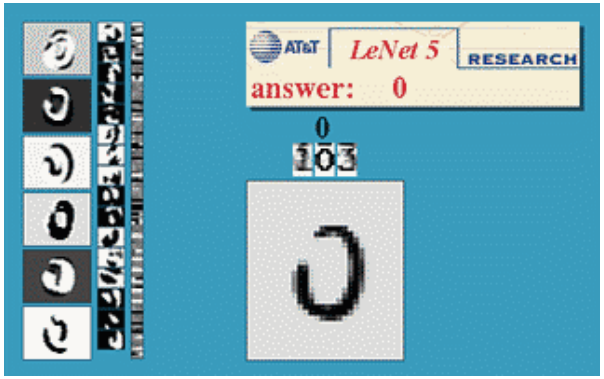
Recognize and generalize patterns dynamically with matrices of probabilistic, weighted neurons

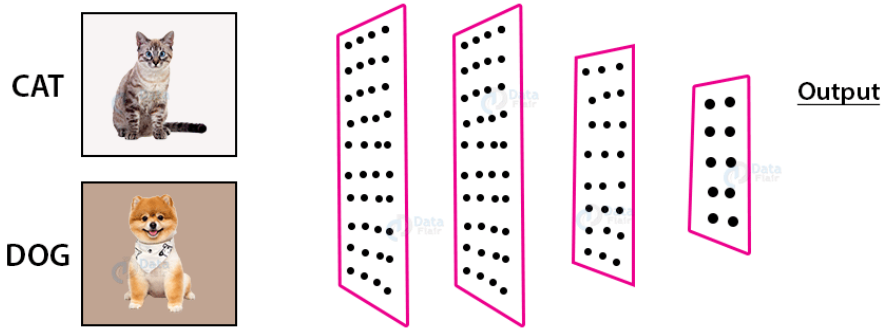
Favored algorithm
Neural networks



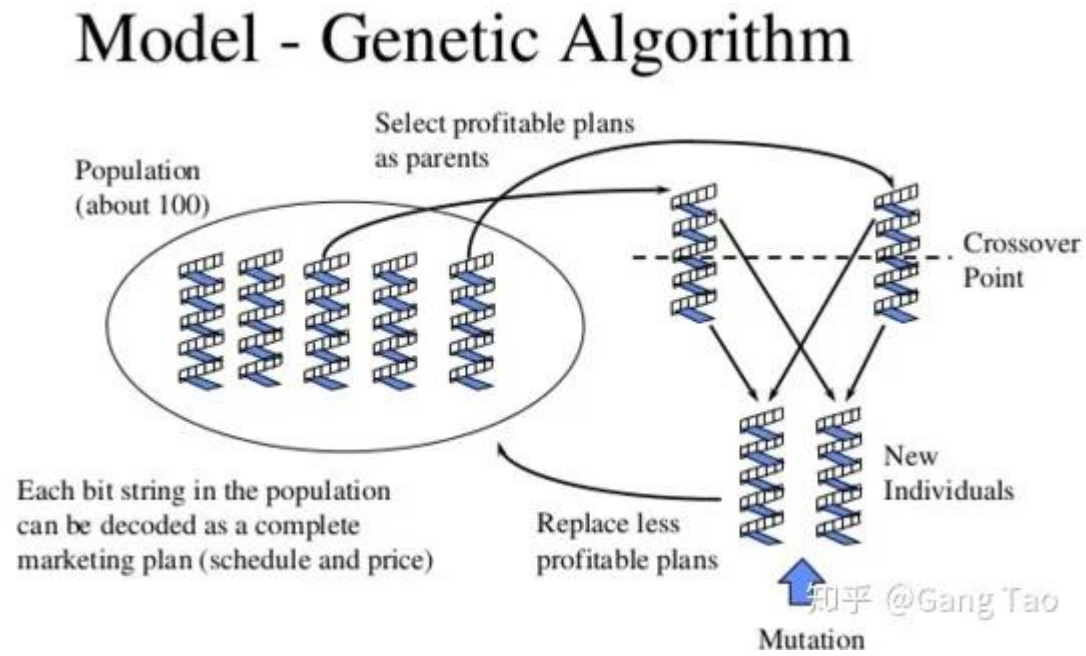


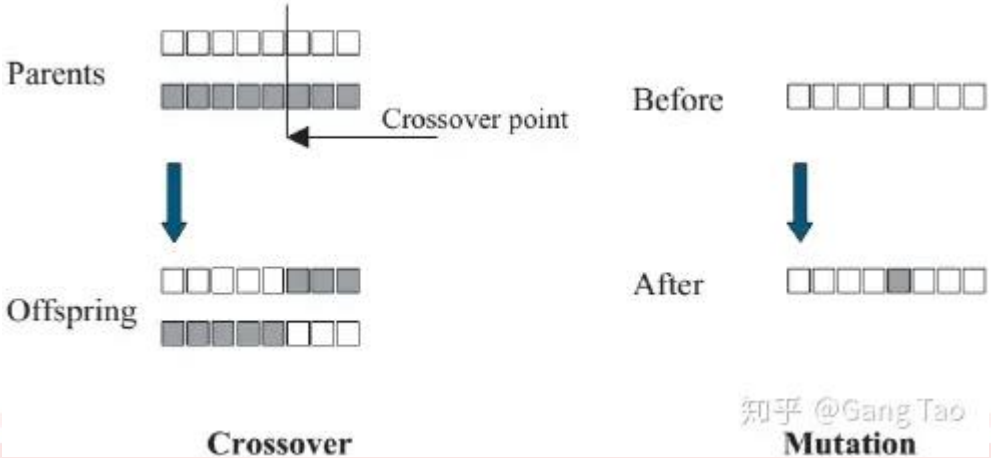
--模仿神经元





4 进化主义





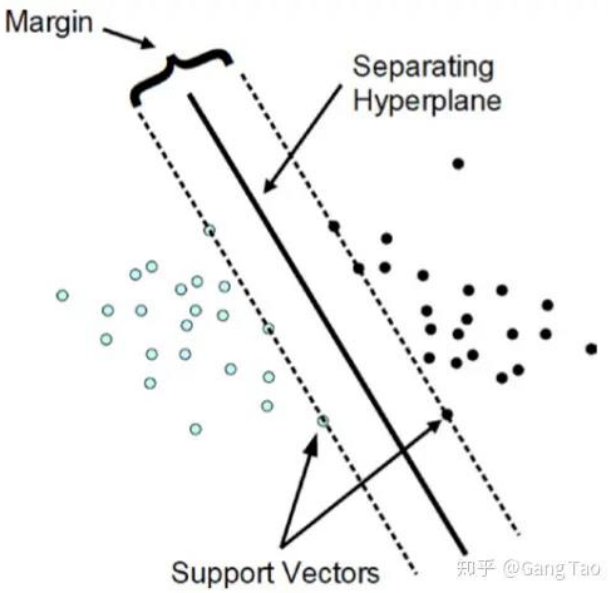
--大自然的学习算法
遗传和变异

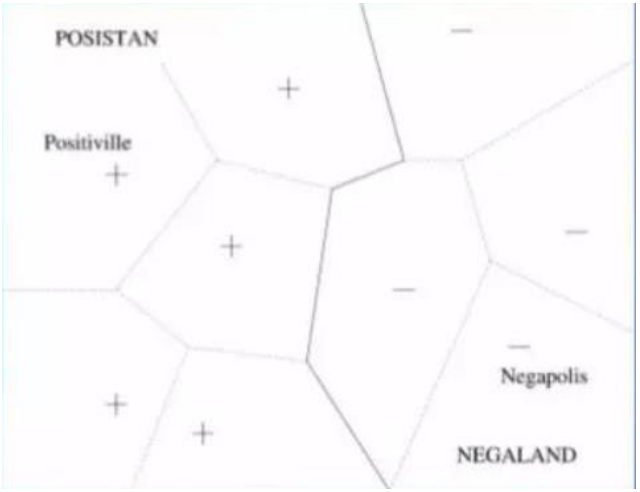
5 类比主义

Analogizers

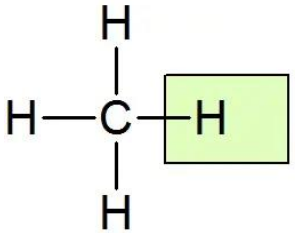
Optimize a function in light of constraints ("going as high as you can while staying on the road")

Favored algorithm
Support vectors

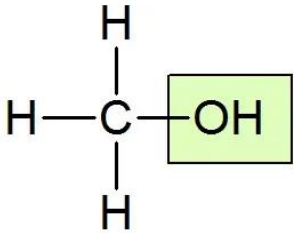




像什么就是什么



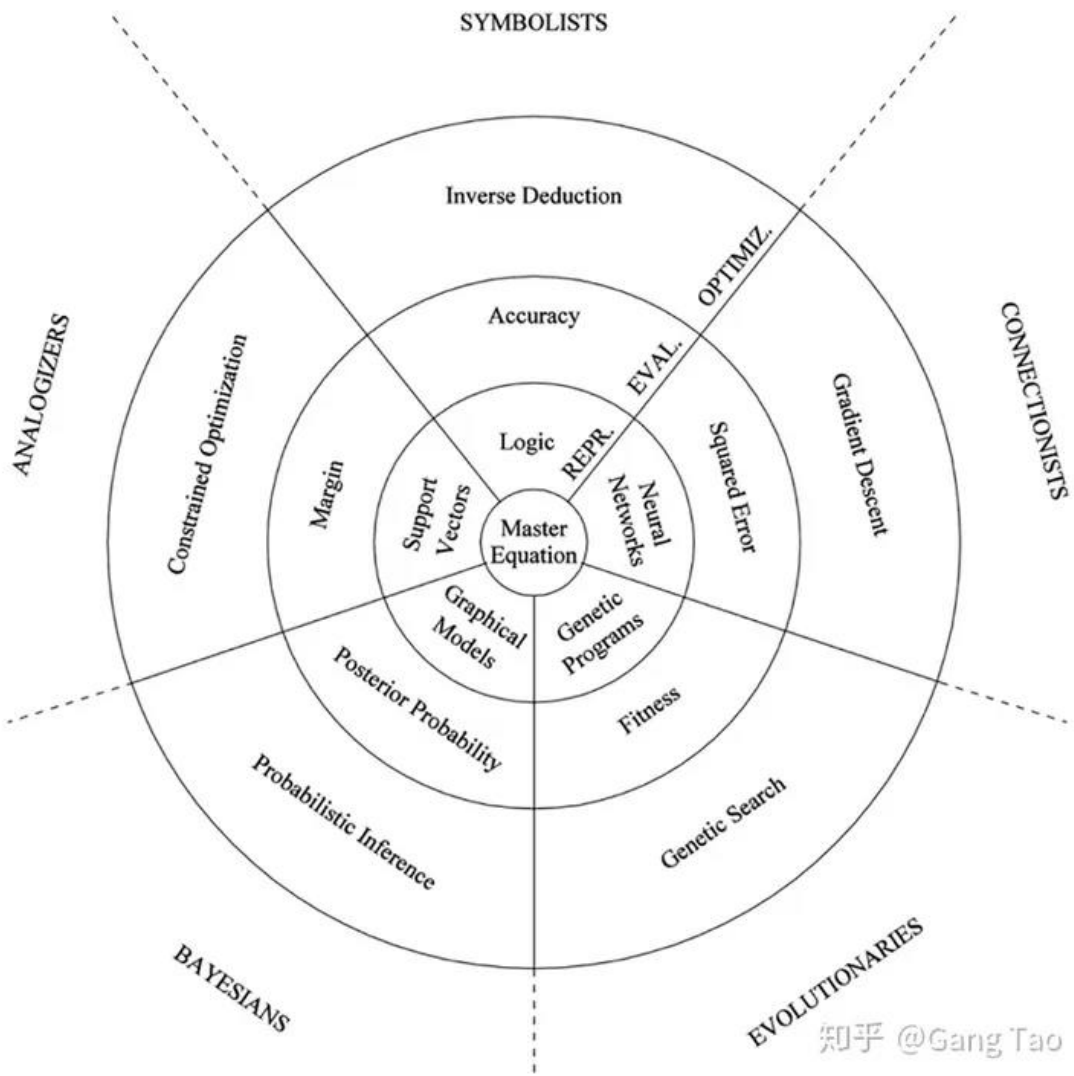
methane



methanol

知乎 @Gang Tao

五大学派的大一统


SYMBOLISTS
Inverse Deduction
OPTIMIZ.
Accuracy
EVAL.
Logic
REPR.
Master Equation
ANALOGIZERS
Constrained Optimization
Margin
Support Vectors
Neural Networks
Squared Error
Gradient Descent
CONNECTIONISTS
Graphical Models
Genetic Programs
Posterior Probability
Fitness
Probabilistic Inference
Genetic Search
BAYESIANS
EVOLUTIONARIES
知乎 @Gang Tao

终极算法

AI school of thought	Challenge	Approach	Key learning algorithm	Application example
Analogizers	Similarity	Kernel machines	Nearest neighbor. Support Vector Machine.	Geographical Information Systems. Handwriting recognition.
Bayesians	Uncertainty	Probabilistic inference	Naïve Bayes	Evidence-based medical diagnostics. Spam filtering.
Connectionists	Credit assignment (Error blaming)	Backpropagation	Neural network (Deep Learning, Autoencoders, Boltzmann Machines)	Image recognition. Speech recognition. Information retrieval.
Evolutionaries	Structure discovery	Genetic programming	Genetic algorithm	Electronic circuit design. Robotics.
Symbolists	Knowledge composition	Inverse deduction	Decision tree/forest	Credit card applications. "Classic" Chess AI.

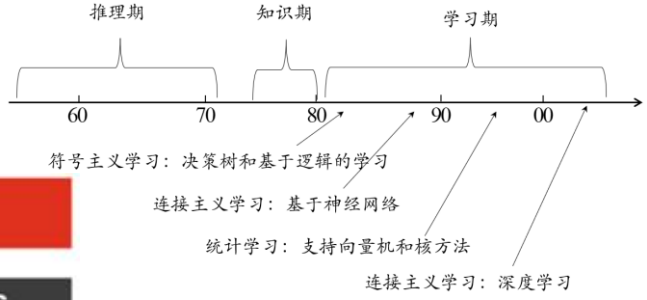
知乎 @Gang Tao

6 行为主义

五大流派+

- 1. 符号主义 --Ch.4 决策树，15 规则学习
- 2. 贝叶斯主义 --Ch.7 贝叶斯分类器, 14 概率图模型
- 3. 连接主义 --Ch.5 神经网络
- 4. 进化主义 --/
- 5. 类比主义 --Ch.6 SVM, 9 聚类
- 6. 行为主义 --Ch.16 强化学习

发展历程



Phases of evolution

1980s

Predominant tribe
Symbolists

Architecture
Server or mainframe

Predominant theory
Knowledge engineering

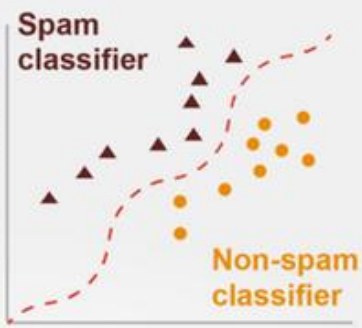
Basic decision logic:
Decision support systems with limited utility

1990s to 2000

Predominant tribe
Bayesians

Architecture
Small server clusters

Predominant theory
Probability theory



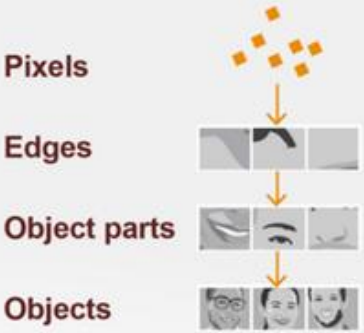
Classification:
Scalable comparison and contrast that's good enough for many purposes

Early to mid-2010s

Predominant tribe
Connectionists

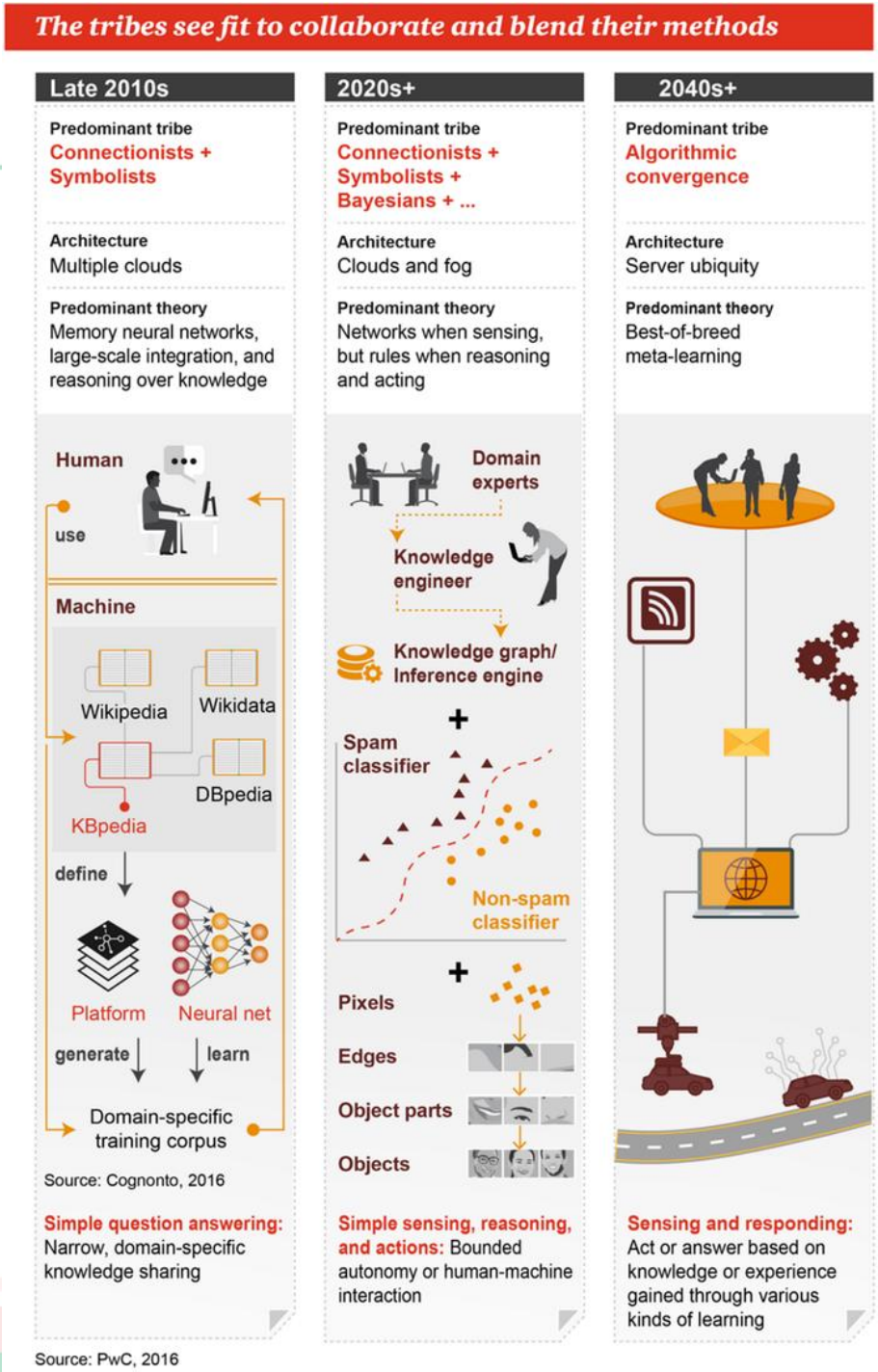
Architecture
Large server farms (the cloud)

Predominant theory
Neuroscience and probability



Recognition:
More precise image and voice recognition, translation, sentiment analysis, etc.

发展历程


The tribes see fit to collaborate and blend their methods
Late 2010s
Predominant tribe
Connectionists + Symbolists
Architecture
Multiple clouds
Predominant theory
Memory neural networks, large-scale integration, and reasoning over knowledge
Human
use
Machine
Wikipedia
Wikidata
DBpedia
KBpedia
define
Platform
Neural net
generate
learn
Domain-specific training corpus
Source: Cognonto, 2016
Simple question answering: Narrow, domain-specific knowledge sharing
2020s+
Predominant tribe
Connectionists + Symbolists + Bayesians + ...
Architecture
Clouds and fog
Predominant theory
Networks when sensing, but rules when reasoning and acting
Domain experts
Knowledge engineer
Knowledge graph/ Inference engine
Spam classifier
Non-spam classifier
Pixels
Edges
Object parts
Objects
Simple sensing, reasoning, and actions: Bounded autonomy or human-machine interaction
2040s+
Predominant tribe
Algorithmic convergence
Architecture
Server ubiquity
Predominant theory
Best-of-breed meta-learning
Sensing and responding: Act or answer based on knowledge or experience gained through various kinds of learning
Source: PwC, 2016

大纲

- 引言
- 基本术语
- 假设空间
- 归纳偏好
- 发展历程
- 应用现状
- 阅读材料

应用现状

- 计算机领域最活跃的研究分支之一：
  - NASA_JPL科学家在Science撰文指出机器学习对科学研究起到越来越大的支撑作用
  - DARPA启动PAL计划，将机器学习的重要性提高到国家安全的高度来考虑
  - 2006年卡耐基梅隆大学宣告成立第一个“机器学习系”，机器学习奠基人之一T.Mitchell教授任系主任。

- 与普通人的生活密切相关：
  - 天气预报、能源勘探、环境监测、搜索引擎、自动驾驶汽车等

应用现状

- 影响到人类社会的政治生活：
  - 2012美国大选期间奥巴马麾下的机器学习团队，对社交网络等各类数据进行分析，为其提示下一步的竞选行动。

- 具有自然科学探索色彩：
  - P.Kanerva在二十世纪八十年代中期提出SDM(Sparse Distributed Memory)模型时并没有刻意模仿脑生理结构，但后来神经科学的研究发现，SDM的稀疏编码机制在视觉、听觉、嗅觉功能的脑皮层中广泛存在，促进理解“人类如何学习”

大纲

- 引言
- 基本术语
- 假设空间
- 归纳偏好
- 发展历程
- 应用现状
- 阅读材料

阅读材料

- [Mitchell, 1997]是第一本机器学习专门教材. [Duda et al., 2001; Alpaydin, 2004; Flach, 2012]为出色的入门读物. [Hastie et al., 2009]为进阶读物, [Bishop, 2006]适合于贝叶斯学习偏好者. [Shalev-Shwartz and Ben-David, 2014]适合于理论偏好者.

- 《机器学习:一种人工智能途径》 [Michalski et al., 1983]汇集了20位学者撰写16篇文章，是机器学习早期最重要的文献. [Dietterich, 1997] 对机器学习领域的发展进行了评述和展望。

阅读材料

- 机器学习领域最重要的国际学术会议是国际机器学习会议(ICML)、国际神经信息处理系统会议(NIPS)和国际学习理论会议(COLT),重要的区域性会议主要有欧洲机器学习会议(ECML)和亚洲机器学习会议(ACML);最重要的国际学术期刊是Journal of Maching Learning Research和Machine Learning.

- 国内不少书记包含机器学习方面的内容，例如[陆汝钤,1996].[李航,2012]是一统计学习为主题的读物. 国内机器学习领域最重要的活动是两年一次的中国机器学习大会(CCML)以及每年举行的“机器学习及其应用”研讨会(MLA).